Compactness of propositional satisfiability

The Open Logic Project contributors

openlogic:propositional-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:compactness (508-512); the proof span 514-519 is the author proof environment that immediately follows it.

Editorial note. The author proof cites no result by label; the completeness and soundness theorems it leans on are external references of this card and are recorded as standard prerequisites instead, so no source text outside the counted span enters the delivered body.

Editorial scope. One original result of the Open Logic Project unit propositional-logic/syntax-and-semantics, counted once: the Compactness Theorem for satisfiability. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Completeness for propositional logic. If a set of sentences is consistent then it is satisfiable; the unit proves this as its completeness theorem.. The first link runs the contrapositive: if Gamma is unsatisfiable it cannot be consistent (558-559).
- Soundness for propositional logic. If a set of sentences is satisfiable then it is consistent; the unit proves this as its soundness theorem.. The contrapositive converts inconsistency into unsatisfiability, both for Gamma itself (558) and again for the finite subset Gamma_0 at the end of the chain (562-563).
- Finiteness of derivations. A derivation is a finite tree of rule applications whose open assumptions form a finite set; hence if Gamma proves two contradictory sentences then some finite subset Gamma_0 of Gamma already proves both.. This unpacks the middle link of the chain (559-561): inconsistency of Gamma means two derivations exist from Gamma, and the finitely many assumptions they use make up the finite subset Gamma_0 that is itself inconsistent.

Proposition 0.1 (Compactness Theorem)**.** Γ *is satisfiable if and only if every* finite *subset* Γ_0 *of* Γ *is satisfiable.*

Proof. Γ is unsatisfiable if and only if it is inconsistent, if and only if some finite subset Γ_0 of Γ is inconsistent, if and only if some finite subset Γ_0 of Γ is unsatisfiable. □